Ankit Jainar

ee24btech11004@iith.ac.in | jainarankit7@gmail.com | +91 8660346551 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	5.51
IKUE INTERNATIONAL PU COLLEGE	Class XII (Karnataka School Examination and Assessment Board)	2023	85.83%
Sri Chaitanya Techno School SINDHANUR	Class X (CBSE)	2021	89.00%

WORK EXPERIENCE

Python Developmental Activity for Netlists & SAIF | Python

Under the guidance of Prof. Amit Acharya, IIT Hyderabad

- Performed Python-based developmental activities for understanding and parsing **netlists and SAIF (Switching Activity Interchange Format) files**.

PROJECTS

Solar PV Telemetry Anomaly & Loss Analysis Engine

Python - Pandas - NumPy - Time-Series Analysis - Anomaly Detection - Energy Analytics

- Built a **Python-based solar PV telemetry analytics pipeline** to detect operational anomalies and identify deviations in expected asset performance from time-series operational data.
- Developed data-processing and anomaly-detection workflows for **solar asset telemetry**, including handling of abnormal readings, data-quality issues, and performance deviations.
- Quantified detected anomalies in terms of **energy and estimated revenue impact**, identifying **2,701.76 kWh of energy loss and 21,614.10 estimated revenue loss** on the test dataset.
- Automated **anomaly detection, loss calculation, and performance reporting** through a modular analytics pipeline designed for recurring asset-performance monitoring.
- Structured the system to support future extension toward **equipment-level root-cause analysis, asset-level performance comparison, and portfolio-scale anomaly monitoring**.

DeepFocus -- Real-Time Computer Vision & Productivity Monitoring

- Built a real-time behavioral monitoring system using **Python, OpenCV, MediaPipe, and Streamlit**, processing facial landmarks from live camera streams.
- Engineered a **temporal state engine** using EAR, head pose, face visibility, and duration-based transitions to detect focus, drowsiness, sleep, distraction, and away states; reduced EAR signal jitter by **~24%**.
- Developed **session analytics, adaptive focus scoring, CSV/JSON logging, and historical dashboards**, with **350+ automated tests** covering behavioral, integration, robustness, and end-to-end scenarios.

Electricity Consumption Forecasting

- Built a **time-series forecasting pipeline** using Pandas and Scikit-learn with Linear Regression, Random Forest, XGBoost, and LSTM models.
- Performed **temporal feature engineering, model evaluation, and comparative forecasting analysis** for electricity-demand prediction.
- Developed a **Streamlit dashboard** for historical analysis, forecast visualization, and model comparison.

Retail Business Analytics Platform

- Built an end-to-end **Python + SQL analytics pipeline** using Pandas, SQLAlchemy, SQLite, and automated ETL workflows for retail transaction data.
- Designed **SQL analytics and ML models** for revenue, CLV, seller performance, customer segmentation, and sales forecasting using K-Means and XGBoost.
- Developed **Power BI dashboards** to visualize KPIs, customer segments, sales trends, and operational insights.

AI Recommendation Platform

- Built an end-to-end **hybrid movie recommendation system** combining TF-IDF content similarity, item-based collaborative filtering, and popularity signals.
- Developed a modular **FastAPI backend** with REST APIs, validation, middleware, logging, exception handling, and ML inference.
- Created a **Streamlit frontend** and hybrid ranking pipeline using normalized weighted scores for personalized recommendations.

SKILLS

Languages: Python, SQL, C, Verilog

Data & Analytics: Pandas, NumPy, SciPy, Scikit-learn, Anomaly Detection

ML/AI: PyTorch, OpenCV, MediaPipe, RAG, Embeddings, LLM APIs

Backend/Deployment: FastAPI, Streamlit, REST APIs, Git

EDA/VLSI: Verilog, Vivado, Netlist/SAIF Analysis

RELEVANT COURSES

Probability & Statistics - Random Processes - Signal Processing - Power System Analysis - Electrical Machines - Electrical Circuits - Differential Equations & Transforms - Machine Learning - Scientific Programming for Electrical Engineering - Data Structures & Applications